Lakshya JEE 2.0 2025

Maths

Basics Mathematics

DPP: 4

Q1 The root of the equation $4^x - 3 \cdot 2^{x+2} + 32 = 0$ are

(A) $1, 2$ (B) $1, 3$

(C) $2, 3$ (D) $0, 2$

Q2 The value of $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+\ldots\ldots\infty}}}}$ is

(A) 3

(B) $2\sqrt{3}$

(C) $\sqrt{42}$

(D) $3\sqrt{2}$

Q3 The complete set of values of k, for which the quadratic equation $x^2 - kx + k + 2 = 0$, has equal roots consists of

(A) $2 + \sqrt{12}$

(B) $2 \pm \sqrt{12}$

(C) $2 - \sqrt{12}$

(D) $-2 - \sqrt{2}$

Q4 If the roots of the equation $12x^2 - mx + 5 = 0$ are in the ratio $2 : 3$, then $m =$

(A) $5\sqrt{10}$

(B) $3\sqrt{10}$

(C) $2\sqrt{10}$

(D) $10\sqrt{5}$

Q5 $x^2 + x + 1 + 2k\left(x^2 - x - 1\right) = 0$ is a perfect square for how many values of k

(A) 2 (B) 0

(C) 1 (D) 3

Q6 Range of the expression $\frac{16x^2-12x+9}{16x^2+12x+9} : (x \in R)$ is-

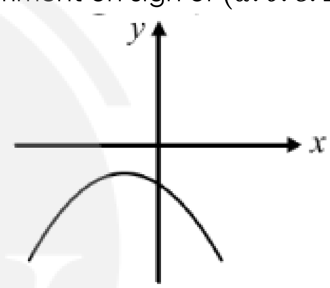
(A) $\left[\frac{1}{3}, 3\right]$

(B) $\left(-\infty, \frac{1}{3}\right]$

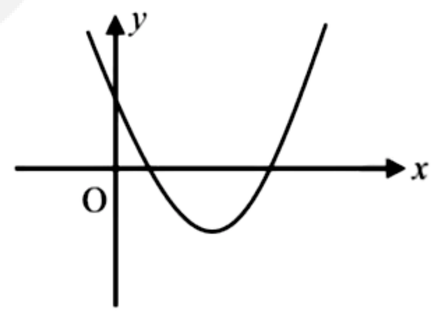
(C) $[3, \infty)$

(D) R

Q7 For the given graph of $\mathrm{f}(x) = ax^2 + bx + c$, comment on sign of $(a.b.c.D)$



Q8 The graph of $y = ax^2 + bx + c$ is shown in figure. Which of the following does **NOT** hold good?



(A) $ab^2c^3 > 0$

(B) $ab^3c^2 < 0$

(C) $ab^3c^5 > 0$

(D) $b^2 > 4ac$

Q9

If value of a for which the sum of the squares of the roots of the equation $x^2-(a-2)x-a-1=0$ assume the least value is-

(A) 2
(B) 3
(C) 0
(D) 1

Q10 Draw graph of $\mathrm{f}(x)=1-\left(x^2+4x\right)$.

Q11 For $x\in[1,5], y=x^2-5x+3$ has

(A) least value of $=-1.5$
(B) greatest value of $=3$
(C) least value $=-3.25$
(D) greater value $=\frac{5+\sqrt{13}}{2}$

Q12 If x is real, then the maximum and minimum values of the expression $\frac{x^2-3x+4}{x^2+3x+4}$ will be

(A) $2,1$
(B) $5,\frac{1}{5}$
(C) $7,\frac{1}{7}$
(D) $2,7$

Q13 The equation $e^{\sin x}-e^{-\sin x}-4=0$ has :

(A) infinite number of real roots
(B) no real roots
(C) exactly one real root
(D) exactly four real roots

Q14 The integer $'k'$, for which the inequality $x^2-2(3k-1)x+8k^2-7>0$ is valid for every x in R, is

(A) 4
(B) 2
(C) 3
(D) 0

Q15 The least integral value of $'k'$ for which $(k-2)x^2+8x+k+4>0$ for all $x\in R$, is :

(A) 5
(B) 2
(C) 3
(D) None of these

Q16 If $x\in R$, then find range of $\frac{x+2}{2x^2+3x+6}$.

Q17 Consider $y=\frac{2x}{1+x^2}$, where x is real, then the range of expression y^2+y-2 is

(A) $[-1,1]$
(B) $[0,1]$
(C) $[-9/4,0]$
(D) $[-9/4,1]$

Q18 Solve the equation:

$$9\left(x^2+\frac{1}{x^2}\right)-27\left(x+\frac{1}{x}\right)+8=0, x\in R$$

Q19 Solve the equation:

$2x^4+x^3-11x^2+x+2=0$

Answer Key

Q1 (C)

Q2 (A)

Q3 (B)

Q4 (A)

Q5 (A)

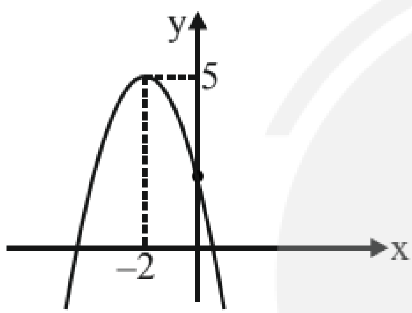
Q6 (A)

Q7 abcD > 0

Q8 (C)

Q9 (D)

Q10



Q11 (B, C)

Q12 (C)

Q13 (B)

Q14 (C)

Q15 (A)

Q16 [-1/13, 1/3]

Q17 (C)

Q18 3 & 1/3

Q19 $x = 1/2,\ 2,\ \left(-3+\sqrt{5}\right)/2,\ \left(-3-\sqrt{5}\right)/2$